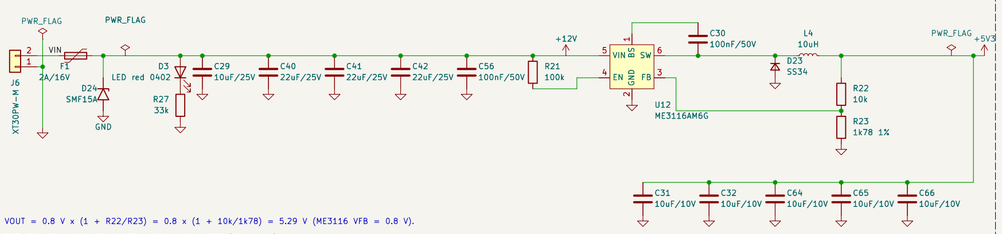
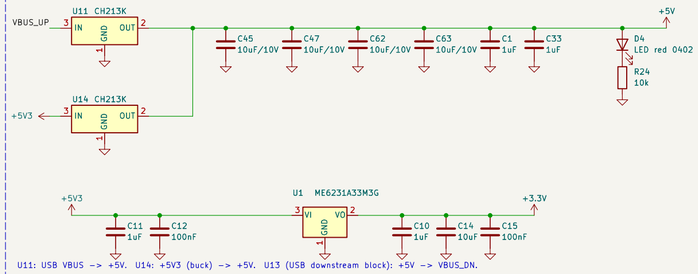
12 V input, 5.3 V buck (ME3116)



VOUT = 0.8 V x (1 + R22/R23) = 0.8 x (1 + 10k/1k78) = 5.29 V (ME3116 VFB = 0.8 V).

D23: Schottky catch diode. EN pulled up to +12V (always on).

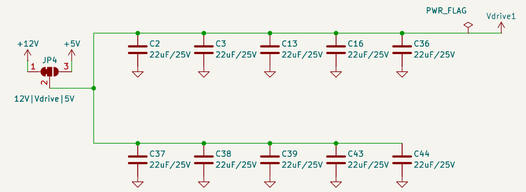
Power path (CH213K ideal diodes), 3.3 V LDO



U11: USB VBUS -> +5V. U14: +5V3 (buck) -> +5V. U13 (USB downstream block): +5V -> VBUS_DN.

+3.3V is made from +5V3 only: the board needs 12 V on J6 to run.

Motor supply Vdrive (JP4 3-pad: 1-2 = +12V bridged by default)

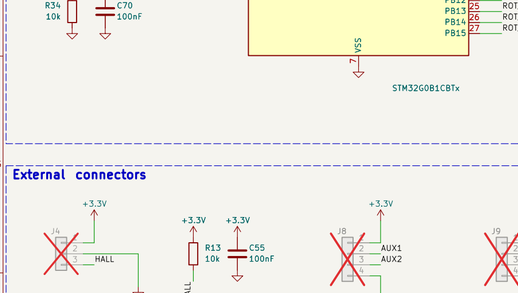


Default: JP4 1-2 bridged (motors on +12V). For 5 V motors: cut the 1-2 trace, then bridge 2-3.

JP4 selects one source only (#70): bridging 1-2-3 would short +12V onto +5V. Clear 1-2 before bridging 2-3.

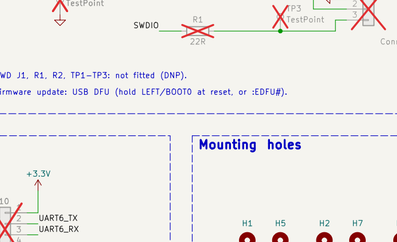
Bulk 10x 22uF/25V shared by both TMC2209 VS pins.

USB-C upstream (to host), ESD



CC1/CC2 5k1 pull-downs: sink (UFP). 22R series + USBLC6 ESD before the hub.

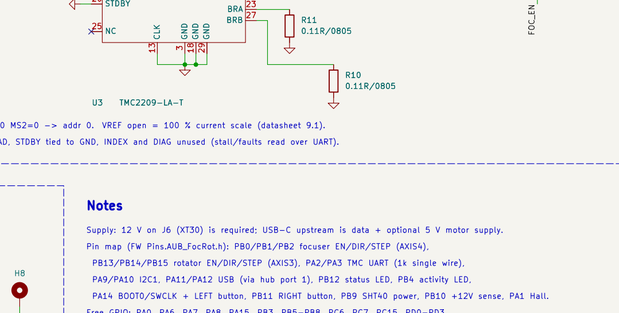
USB 2.0 hub CH334P (3.3 V supply)



Port 1 -> MCU USB, port 2 -> downstream USB-C, ports 3/4 unused. Y2 has built-in load caps.

V5 tied to VDD33 (3.3 V supply mode). RESET#/CDP and PGANG open.

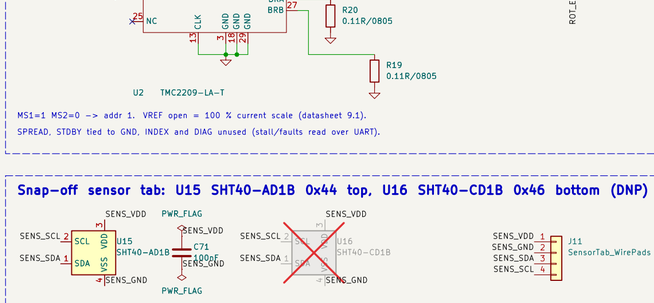
USB-C downstream port



U13 (CH213K) feeds VBUS_DN from +5V. 22R series + USBLC6 (U9) ESD at J7.

CC1/CC2 56k pull-ups to +5V: source Rp, default USB power (500 mA).

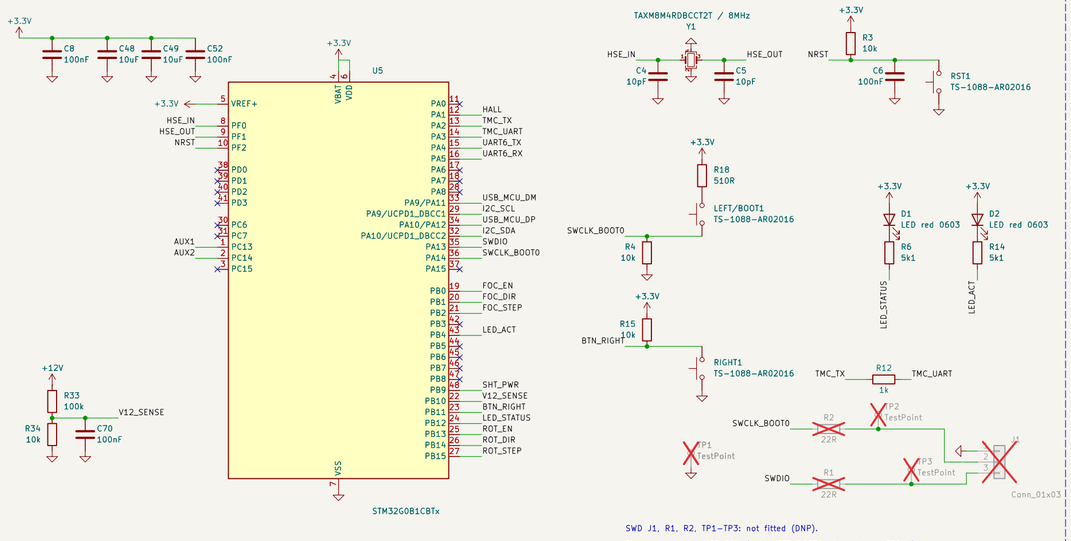
I2C: EEPROM 24AA64, SHT40, IMU MPU-6000



EEPROM 0x50, WP low. SHT40-BD1B on board (U8, fixed 0x45). VDD from PB9 via R32; sensor tab via J9: U15 0x44, U16 0x46 (DNP).

MPU-6000: AD0=0, CS=1 (I2C mode) -> 0x68. INT not connected.

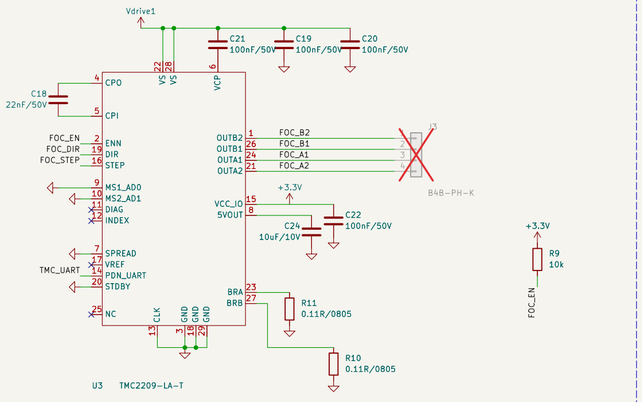
MCU STM32G0B1CBT6



SWD J1, R1, R2, TP1-TP3: not fitted (DNP).

Firmware update: USB DFU (hold LEFT/BOOT0 at reset, or :EDFU#).

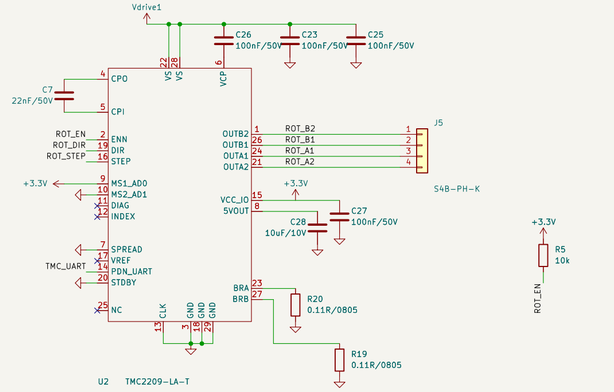
Focuser driver TMC2209 (AXIS4, UART addr 0)



MS1=0 MS2=0 -> addr 0. VREF open = 100 % current scale (datasheet 9.1).

SPREAD, STDBY tied to GND, INDEX and DIAG unused (stall/faults read over UART).

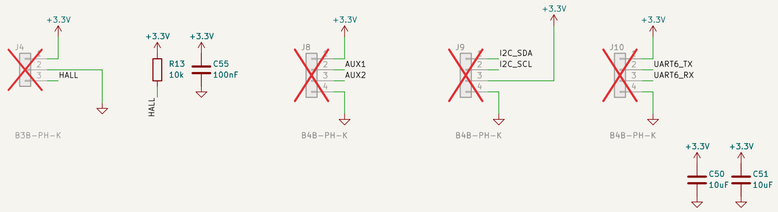
Rotator driver TMC2209 (AXIS3, UART addr 1)



MS1=1 MS2=0 -> addr 1. VREF open = 100 % current scale (datasheet 9.1).

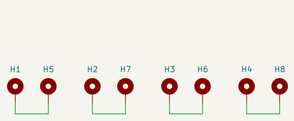
SPREAD, STDBY tied to GND, INDEX and DIAG unused (stall/faults read over UART).

External connectors



J4 Hall (home), J8 AUX PC13/PC14, J9 I2C, J10 USART6 PA4/PA5. All on +3.3V.

Mounting holes



M2 pads H1-H4 tied to M3 pads H5-H8 (not GND).

Notes

Supply: 12 V on J6 (XT30) is required; USB-C upstream is data + optional 5 V motor supply.

Pin map (FW Pins.AUB_FocRot.h): PB0/PB1/PB2 focuser EN/DIR/STEP (AXIS4),

PB13/PB14/PB15 rotator EN/DIR/STEP (AXIS3), PA2/PA3 TMC UART (1k single wire),

PA9/PA10 I2C1, PA11/PA12 USB (via hub port 1), PB12 status LED, PB4 activity LED,

PA14 BOOT0/SWCLK + LEFT button, PB11 RIGHT button, PB9 SHT40 power, PB10 +12V sense, PA1 Hall.

Free GPIO: PA0, PA6, PA7, PA8, PA15, PB3, PB5-PB8, PC6, PC7, PC15, PD0-PD3.

Rev 2.0: net names, ESD on J7 (U9), Y2 caps, SHT31 decoupling, EEPROM WP, LED and +5V3 values.

Rev 2.1: R32 47R in series with the SHT31 supply (PB9), Y2 load caps C67/C68 removed, +12V sense R33/R34/C70 on PB10, JP4/JP5 -> 3-pad JP4, F1 2 A polyfuse + D24 SMF15A TVS on the 12 V input.

Rev 2.2: U8 SHT31 -> SHT40-BD1B (fixed 0x45, C69 moved), snap-off SHT40-AD1B sensor tab (U15, C71, J11, MB1) on the left board edge.

Rev 2.3: D1/D2 LEDs 0603 (C2286), C34/C35 USB shield caps 4.7nF/50V, backup SHT40-CD1B U16 (0x46, DNP) on the bottom of the sensor tab, top-edge GND vias moved in for panel tabs.

Snap-off sensor tab: U15 SHT40-AD1B 0x44 top, U16 SHT40-CD1B 0x46 bottom (DNP)

Tab 4 x 10 mm on the left board edge, held by one mouse-bite row (MB1: 4 NPTH 0.6 mm, 1.0 mm pitch). No copper crosses the snap line. U16 = backup on the bottom, two vias next to C71 (shared decoupling), DNP; fitting it makes the assembly double-sided.

Snap off, then wire to J9: SDA -> 1, SCL -> 2, VDD -> 3, GND -> 4. Pads: VDD and GND on top, SDA and SCL on the bottom. No pour.

Schematic reorganised 2026-11-05 (netlist unchanged)

STM32G0B1 + 2x TMC2209, USB hub, IMU, SHT40 + snap-off SHT40 tab

AstroUniBoard

Sheet: /

File: astrouniboard.kicad_sch

Title: AstroUniBoard — focuser / rotator controller

Size: A2	Date: 2026-11-09	**Rev: 2.3**
KiCad E.D.A. 10.0.0		Id: 1/1